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Benjamin Cabanes, Quentin Plantec, Pascal Le Masson, Benoît Weil

► To cite this version:

Benjamin Cabanes, Quentin Plantec, Pascal Le Masson, Benoît Weil. Basic or applied research in university–industry collaborations? An analysis of research orientations in knowledge creation partnerships. *European Management Review*, 2024, 10.1111/emre.12696 . hal-04742889

HAL Id: hal-04742889

<https://hal.science/hal-04742889v1>

Submitted on 24 Mar 2025

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Basic or applied research in university–industry collaborations? An analysis of research orientations in knowledge creation partnerships

Cabanes, B., Plantec, Q., Le Masson, P. & Weil, B.

ABSTRACT

In a knowledge-based economy, university–industry collaborations play a pivotal role in driving innovation and economic growth. This study investigates the research orientations of these collaborations, focusing on the balance between basic and applied research. Using data from 631 collaborative projects funded by the French Cifre PhD program, we find that most projects emphasize basic research (61%), contradicting the common expectation that industry-led partnerships primarily focus on applied research. Importantly, our analysis reveals that this preference for basic research is not contingent on traditional contextual factors such as scientific discipline, company size, or university type. These findings challenge established assumptions in the literature and suggest that even in industry-sponsored research, fundamental inquiry is critical in supporting long-term innovation strategies. This paper contributes to the understanding of research orientations in university–industry collaborations and offers practical suggestions for enhancing these partnerships and informing research policy. Additionally, we outline a research agenda for further exploration of the factors influencing research orientations, the impact of early-stage industry engagement, and the role of policies in shaping collaboration outcomes.

KEYWORDS

University–industry collaboration, knowledge creation, knowledge transfer, research orientations, basic and applied research

INTRODUCTION

In a knowledge-based economy (Powell & Snellman, 2004), university–industry collaborations are considered a key element and have gained tremendous importance in the last two decades (Etzkowitz & Leydesdorff, 2000; Gibbons et al., 1994; He et al., 2021). These collaborations, which notably expand universities’ traditional teaching and research missions (Compagnucci & Spigarelli, 2020; Fleming et al., 2019; Laredo, 2007), have been strongly encouraged by policymakers and the development of technology transfer offices (Miller et al., 2018).

University–industry collaborations include various forms of partnerships and interactions with the goal of taking advantage of the particular strengths of academia and industry to promote research, innovation, economic growth, and societal impact. We can distinguish at least two models of collaboration. The first model, which has been extensively documented in the literature, is based on *knowledge transfer* (Chai & Shih, 2016; Rajalo & Vadi, 2017; Siegel et al., 2003; Terán-Bustamante et al., 2021). This collaboration model includes technology transfer and commercialization, such as patent licensing, entrepreneurship support to launch startup ventures based on academic research findings, consulting and expertise to provide technical assistance, or collaborative facilities to share access to specialized resources. The second model is based on *knowledge creation*. This type of collaboration — which has received less attention than collaborations based on knowledge transfer (Wit-de-Vries et al., 2019) — includes joint research projects, industry-sponsored research, industry-academia partnerships, university–industry consortia, and university–industry cooperative research centres; this latter kind of collaboration, representing the main focus of this study, is not based on the transfer of already produced knowledge but rather aims to cocreate new scientific knowledge (Siegel et al., 2010; Tijssen, 2018).

Collaborating to create new knowledge implies sharing objectives, expectations or, in other words, the same *research orientation* for a given project. Here, research orientation refers to the strategic direction and focus taken by researchers or collaborative groups in their pursuit of knowledge creation, characterized by the intent concerning and approach to either basic or applied research or a blend of both, known as use-inspired basic research (e.g., Stokes, 1997). However, we know very little about the research orientation on which university–industry collaborative research projects rely. This lack of knowledge is partly due to the fact that academia and industry function within separate institutional contexts that are defined by their own sets of norms and incentives, which may sometimes be at odds with each other’s efforts (Bjerregaard, 2010; Dasgupta and David, 1994; Sauermann and Stephan, 2013). The differences in research orientations often lead to tensions between (1) researchers in universities and public laboratories, who are more prone to pursue a more basic research approach, and (2) researchers in corporate R&D departments, who are more prone to taking a more applied research approach (Balconi et al., 2010; Bentley et al., 2015; Godin, 2006, 2011; Gulbrandsen & Kyvik, 2010). Indeed, if research questions are prescribed by the corporate business strategy, then researchers in industrial laboratories may produce new knowledge for the immediate needs of the company, and research investment is likely to be profitable through the commercialization of the company’s products and technologies. In contrast, basic research cannot be motivated by the quest for immediate results or subject to business pressures. While the research orientation can be based on a more mixed approach, combining, for example, basic research with a consideration of the usage of the produced knowledge sounds like a more exceptional case than the norm in university–industry collaborations (Stokes, 1997).

Therefore, this research aims to explore the different types of research orientations in university–industry collaborations and the extent to which they are chosen due to certain characteristics of the projects. While the literature has extensively discussed the

performance and success factors of university–industry collaborations, in-depth understanding of the initial objectives and goals of such collaborations remains lacking. In particular, numerous studies have shown that the lack of alignment in research orientation intentions between universities and industries constitutes a critical barrier to collaboration (Bruneel et al., 2010; Hemmert et al., 2014; Tartari et al., 2012); however, previous research has largely been limited to the study of scientific yields and the evaluation of collaboration outcomes (Callaert et al., 2015; He et al., 2021; Perkmann et al., 2013; Sauermann and Stephan, 2013; York et al., 2016), despite calls to improve our understanding research strategies, goals, practices and objectives in the context of this type of collaboration (e.g., Callaert et al., 2015; Tijssen, 2018; Plantec et al., 2023). Indeed, as highlighted by Perkmann et al. (2021, p. 9), we must learn more about *“the direction of research, beyond the question of whether engagement pushes academics towards more applied research, [...] Notably, it would be interesting to know whether academic engagement propels scientists to be more exploratory in their research – moving into new areas and tackling new problems”*.

To help fill this gap, it is necessary to improve our understanding of how the university–industry relationship may (or may not) be related to different research orientations and to identify the characteristics that are associated with the choice of a specific research orientation in the context of a given collaboration. Understanding the nature of research orientations in the formation of university–industry collaborations is crucial due to their significant implications for innovation, economic growth, and societal advancement (Perkmann et al., 2021). Exploring this aspect of collaboration can contribute to uncovering the mechanisms through which research orientations influence the initiation and structure of these partnerships. This is particularly important in a knowledge-driven economy, where the seamless integration of basic and applied research is essential for addressing complex global challenges, fostering innovation, and ensuring

the competitiveness of industries. Additionally, a deeper understanding of how research orientations affect collaborative ventures can guide policymaking, enhance the management of university–industry relations, and optimize the strategic alignment of research agendas between academia and industry, thereby maximizing the societal and economic benefits of research investments.

For this purpose, we explore the case of the Cifre programme of the French National Association for Research and Technology (ANRT). The Cifre programme is a type of knowledge creation-oriented university–industry collaboration that allows firms to finance PhD students’ efforts to complete a three-year research project through university–industry collaboration. One key aim of exploring the Cifre programme is that this programme covers all types of scientific disciplines (ranging from science, technology, engineering, and mathematics (STEM) disciplines to the social sciences) and companies (including startups, small and medium-sized enterprises (SMEs), and large groups) in every sector as well as in all regions of France. Our research is based on a study of 631 collaborative research projects, in which context we analyse fine-grained data that is closely related to the notion of contractual collaborative agreement (e.g., in terms of type of industry, discipline, financial amount, and the amount of time spent by the team in each facility) and define the project research orientation through manual double-blinded coding. The Cifre programme exhibits characteristics that are similar to those of other university–industry programmes in other countries, especially in Europe, where this type of collaborative PhD has become highly important (e.g., Salimi et al., 2016).

Due to the exploratory nature of this research, we rely primarily on a descriptive approach, as has been the case in previous studies on university–industry relationships (e.g., Cohen et al., 2002). In contrast to the belief that actors in industry are willing to rely on more applied research, our findings reveal that most university–industry partnerships focus on the creation of new scientific knowledge regardless of its immediate

practical applications. Specifically, 61% of projects in the Cifre context prioritize basic research in line with academic research conventions across various scientific disciplines, university types, and company sizes. We also observe that companies that are new to the programme tend to engage more in use-inspired basic research, thus offering room for further exploration of this element. This overall shift towards academic conventions highlights a nuanced approach to collaboration, which values fundamental understanding as a key component of industry-related innovation. While acknowledging the distinction between academic and industrial logics, our findings highlight that cultural differences between academic and corporate research do not necessarily impede collaborative efforts aimed at generating new knowledge. These collaborations often focus on exploring emerging scientific fields and, in the context of research partnerships, tend to adopt academic conventions that prioritize orientations centred on basic research.

The remainder of this article is organized as follows. First, we present a literature review on university–industry collaborations and research orientations. Second, we present our empirical context and the data collection and analysis processes. Third, we present our empirical results. Finally, we conclude this work with a discussion of our contributions and a presentation of potential future research directions.

LITERATURE REVIEW

Both universities and firms have benefits and motivations related to university–industry collaborations (Bishop et al., 2011; D’Este & Perkmann, 2011; Lee, 1996, 2000; Perkmann & Walsh, 2009; Siegel et al., 2003). For universities, collaborating with a firm can support development by fostering relationships with knowledge users and facilitating technology transfer (Etzkowitz & Goktepe, 2005; Perkmann et al., 2013), which also includes improving funding for research projects and sharing knowledge and resources (D’Este & Perkmann, 2011; Garcia et al., 2019). For firms, close collaboration with universities allows

them to gain access to new knowledge, innovative technology, state-of-the art expertise, research networks and facilities (Ankrah & Al-Tabbaa, 2015). Therefore, understanding university–industry collaborations is important for management research and innovation policy.

As evidenced by the large number of systemic literature reviews, university–industry collaborations include several motivations and objectives based on different practices and organizations (Ankrah et al., 2013; Miller et al., 2018; Sjöo & Hellström, 2019; Skute et al., 2019; Vick & Robertson, 2018). The objectives may concern the resolution of technological problems, the sharing of scientific resources, the commercialization of university-based technologies, the creation of new knowledge or the creation of new businesses. These objectives are based on formal or informal interactions such as licensing agreements, royalty agreements, research partnerships and consulting.

In recent decades, universities and firms have increasingly been encouraged to collaborate on research and development activities, guided by the principles of Mode 2 knowledge production (Gibbons et al., 1994), the triple-helix model (Etzkowitz & Leydesdorff, 1997, 1999, 2000), and the third mission of the university (Compagnucci & Spigarelli, 2020; Laredo, 2007). These new approaches aim to move beyond simple knowledge transfer as described by the linear model of innovation (Godin, 2006, 2011). The linear model posits that innovation follows a sequential process, beginning with basic research, progressing through applied research, and culminating in product or service development by a company, which is then introduced to the market (Gulbrandsen & Kyvik, 2010; Henard & McFadyen, 2005; Narayanamurti & Odumosu, 2016).

Gibbons et al. (1994) argued that scientific knowledge production has shifted from a Mode 1 paradigm, characterized by the strict separation of basic and applied research, to a Mode 2 paradigm based on the convergence of research orientations. While Mode 1 relies on stabilized disciplines and the distinction between fundamental and applied

research, Mode 2 is transdisciplinary, with a constant interplay between the fundamental and the applied, and between theory and practice. Mode 2 represents a shift towards inquiry focused on contextualized results. Similarly, the concepts of the “Triple Helix of university-industry-government relations” (Etzkowitz & Leydesdorff, 1997, 1999, 2000) and the third mission of the university (Compagnucci & Spigarelli, 2020; Laredo, 2007) emphasize the dynamic, interactive relationships among universities, industry, and government in fostering innovation and economic development. Integrating knowledge creation, including basic and applied research, allows these entities to collaborate more effectively to drive innovation and societal progress.

University–industry collaborations: Knowledge transfer or creation?

Collaboration based on knowledge transfer is a process that allows scientific discoveries, knowledge, and intellectual assets to flow from creators, such as universities and research institutions, to both public and private users. The aim of such collaboration is to convert inventions and scientific knowledge into new products and services that contribute to societal needs (de Wit-de Vries et al., 2019; Geuna & Muscio, 2009; Lundquist, 2003; Miller et al., 2018). Knowledge is first produced by the university independently of application issues, and then, this knowledge is transferred, absorbed and used by the private sector to address innovation and business needs (Ankrah et al., 2013; Miller et al., 2018; Sjöö & Hellström, 2019; Skute et al., 2019; Vick & Robertson, 2018). In this context, university–industry collaborations are a way for firms to obtain expertise that cannot be generated in-house through knowledge transfer (Fernandes & O’Sullivan, 2023; Rajalo & Vadi, 2017; Siegel et al., 2003; Terán-Bustamante et al., 2021).

There are at least two ways to transfer scientific knowledge: the direct and indirect placement of products and services in the market (Carlsson & Fridh, 2002; Doganova, 2013; Wright et al., 2008). University spin-offs are a direct mechanism, representing the

entrepreneurial route to the commercialization of public research (Berbegal-Mirabent et al., 2015). Licensing agreements on academic patents, royalty agreements, access to research networks and consulting are indirect mechanisms through which firms and universities collaborate on a win–win basis. Knowledge transfer, also called technology transfer, is managed by a TTO, which is a specialized organizational structure within the university dedicated to managing supply and demand in terms of knowledge transfer. The main functions include intellectual property management, licencing and technology commercialization and entrepreneurship support (Etzkowitz & Goktepe, 2005; Macho-Stadler et al., 2007; Miller et al., 2018).

Knowledge transfer channels between universities and industry depend on the context and can be influenced by a myriad of factors, including the industrial sector, research orientations, scientific disciplines, and the typology of organizations (Bekkers & Bodas Freitas, 2008). For example, collaborations with universities are particularly important in sectors where technology is rapidly evolving, as companies seek to stay active across multiple technological trajectories (Belderbos et al., 2004; Marsili, 2001). The intensity of collaborations also varies by scientific discipline, with a higher frequency observed in the fields of chemistry, biotechnology, engineering, and information technology (Martinelli et al., 2008; Schartinger et al., 2002). In addition, collaborations also vary depending on the type and size of the organisation. Santoro and Chakrabarti (2002) highlight that firms of varying sizes and sectors engage in different forms of interaction with universities to achieve specific objectives, such as skill development or addressing challenges in both core and non-core technological areas. Additionally, several studies indicate that large firms and those with significant R&D investments are more likely to possess the absorptive capacities necessary for learning from and collaborating with universities (Cohen et al., 2002; Fontana et al., 2006). Moreover, firms with multitechnology strategies often find it crucial to employ diverse approaches for

systematically and autonomously accessing and developing technologies (Granstrand et al., 1997).

Collaboration based on knowledge creation is a process that allows the creation of new scientific knowledge by building on a common research programme and addressing specific research questions that are useful to all parties (Engestrom et al., 2003; Hong & Olander, 2010; Inkpen, 1996). Unlike those based on knowledge transfer, the ambition of this type of collaboration is not to commercialize existing knowledge but to create new knowledge. These collaborations can take various forms, such as research partnerships, joint venture research projects, university–industry consortia, or university–industry cooperative research centres (Hagedoorn et al., 2000; Oliver, 2022; Mannak et al., 2019). These collaborations often involve interdisciplinary teams working on projects funded by grants, contracts, or collaborative agreements. These collaborations can lead to scientific findings, new knowledge creation, new technology development, inventions, patent creation and innovations.

Research orientations in knowledge creation collaborations

Collaborative research, where academic researchers collaborate with companies to conduct knowledge creation to address industry-specific challenges, is a common activity for both science-based organizations and research-intensive universities. However, these collaborations also face theoretical and practical barriers and challenges related to various aspects of partnership formation, dynamics, goals, effectiveness, and outcomes. Bruneel et al. (2010) highlighted the following two types of barriers limiting university–industry collaborations: barriers related to conflicts over IP and barriers related to differences in the research orientations of industry and universities.

Research orientations refer to the distinction between basic and applied research (Busch, 1945; Gulbrandsen & Kyvik, 2010; Henard & McFadyen, 2005; Hottenrott, 2012).

Applied research focuses on addressing specific practical problems, the results have practical and commercial value, and the research is carried out mainly by industry. Research questions are prescribed by corporate business strategies; researchers in industrial laboratories produce new knowledge for the immediate needs of companies, and research spending is likely to be profitable through the commercialization of the company's products and technologies. In contrast, basic research is not motivated by the quest for immediate results or subject to business pressures. Basic, or fundamental, research is performed without thought of practical ends and is driven by a quest to understand nature and its laws. As a result, researchers in universities and public laboratories are motivated primarily by basic research, while researchers in corporate R&D departments focus on applied research (Balconi et al., 2010). Academic researchers create new fundamental knowledge and publish their results in scientific journals, while industrial scientists produce applied knowledge in the form of products and manufacturing processes (Partha & David, 1994; Tartari et al., 2012). Given the two different systems of knowledge creation, collaborations between universities and industries are likely to be plagued by conflicts due to a weak alignment of research orientations. According to Garcia et al. (2019, p. 350), academia and industry operate in distinct institutional environments characterized by norms and incentives that can conflict with one another. As a result, we are faced with a paradox.

On the one hand, many studies highlight the distinction between academic and commercial logic as two separate institutional frameworks that can complicate collaborative efforts (Bjerregaard, 2010; Dasgupta and David, 1994; Sauermann and Stephan, 2013). A lack of alignment in research orientations hinders collaboration, as it generates contradictory objectives and expectations on various aspects of collaborative research projects: How can common research questions be shared? How can the parties agree on a common research program? Should the project be limited to a specific research domain? Which research methodologies should be chosen? (Bruneel et al., 2010; Hemmert

et al., 2014; Tartari et al., 2012). These objectives and expectations have been even more difficult to reconcile since the 1980s and 1990s, and industrial companies have largely abandoned fundamental research, thereby closing their main industrial research laboratories. This phenomenon is even characterized as a “*research bloodbath*” by historian Robert Buderer (2000).

On the other hand, other studies consider that research orientations are not obstacles to joint research programmes (Atta-Owusu et al., 2021; Bjerregaard, 2010; Callaert et al., 2015; He et al., 2021; Perkmann et al., 2013; Sauermann and Stephan, 2013; York et al., 2016). According to Bjerregaard (2010), institutional logics are different but complementary. Callaert et al. (2015) suggested that combining scientific and more entrepreneurial activities at the academic scientist level is feasible. By exploring university–industry collaborations in research centres, Lindl et al. (2013) identified four forms of collaboration based on knowledge creation. The distanced form of collaboration is dominated by university logic, in which the industry provides initial funding and ideas, enabling the academic partner to conduct its research independently. The results, produced at the end of the knowledge creation process ultimately benefit the industry. In contrast, the specified form of collaboration, driven by industry logic, focuses on developing and improving industry products. In the developed and translational forms of collaboration, both university and industry logics are influential, fostering a more integrated approach where each party contributes to and benefits from the collaboration. However, developed and translational forms of collaboration differ in one key aspect. In developed collaborations, there is a single, integrated research process in which all parties actively participate. In contrast, in translational collaborations, two interdependent research processes run in parallel. The university controls one process, focusing on research with the aim of publishing papers, while the industry controls the other, concentrating on research aimed at product development and advancing the company. Furthermore,

academic engagement—defined as “*knowledge-related collaboration by academic researchers with nonacademic organizations*” (Perkmann et al., 2013, p. 423)—is positively associated with academics’ subsequent scientific productivity (Bikard et al., 2019; Perkmann et al., 2021). For academics, collaboration with industry “*facilitates interactive learning, which in turn indirectly benefits scientific production by generating new ideas and motivating new research projects*” (Perkmann & Walsh, 2009). Finally, the difference between these two institutional logics may diminish over time, facilitating the convergence of objectives and expectations during research collaboration (Sauermann and Stephan, 2013; York et al., 2016).

Taken together, these contrasting arguments suggest that our understanding of research orientations and their influence on university–industry collaboration formation and implementation is still incomplete. As such, the main objective of the present study is to address the following research questions: (1) What is the nature of research orientations in university–industry collaborations focused on knowledge creation? (2) What is the influence of context on research orientations? (3) What are the relationships between research orientations and forms of collaboration (i.e., distanced, specified, or developed)?

METHODS

Given that there is limited existing knowledge and understanding of the subject, we opt for an exploratory research approach based on a qualitative method (Denzin & Lincoln, 2011; Glaser & Strauss, 1967; Yin, 2003). This approach is appropriate because there is limited existing knowledge or understanding of the subject. Furthermore, our goal is to investigate a topic in a preliminary and open-ended manner to explore and generate insights and hypotheses rather than to test specific hypotheses or theories. An exploratory research approach is valuable for identifying trends, patterns, or relationships that may warrant further investigation in more structured studies (Kumar, 2018). This approach is

also explained by the extreme and unique nature of our case, with the aim of revealing new phenomena that have not yet been addressed or analysed in the literature (Eisenhardt & Graebner, 2007; Siggelkow, 2007). In this research, we decide to focus on university–industry collaborations based on PhD collaborative programmes. PhD collaborative programmes, also called industrial PhDs (Grimm, 2018), refer to research projects that involve a university, a firm and a PhD candidate for a duration of 3-4 years (Salimi et al., 2016). During this time, all parties work together to meet common expectations and cocreate new scientific knowledge.

On the basis of our research question, the choice to study joint PhD projects is justified for at least four reasons. First, this choice ensures that we study collaborative research projects on the basis of knowledge creation. In PhD collaborative programmes, doctoral students are expected to defend a doctoral thesis by demonstrating the production of new knowledge and the achievement of scientific results. Consequently, this situation ensures that we exclude collaborations founded on consulting missions or a simple transfer of already existing knowledge. Second, the study of PhD collaborative programmes allows us to study long-term collaborations based on the reciprocal commitment of both parties (university and industry). Such collaboration implies the sharing of means and the pooling of resources for the realization of the project (knowledge, instruments, data, observations, materials, finance, etc.) and takes place over a period of at least 3 years. Thus, communications or employer brand development operations cannot be considered forms of collaboration. Third, the purpose and objectives of each research project are determined by the firm and the university. Therefore, each collaboration should be unique, and we should be able to highlight a variety of original research projects on the basis of different research orientations. Fourth, university–industry doctoral projects are classic and widely used mechanisms for organizing university–industry collaborations in Western countries such as France, Denmark, the United Kingdom, Germany, Sweden, Spain, the USA and

Australia (Grimm, 2018; Plantec et al., 2023; Salimi et al., 2016). In this way, our results are easier to generalize since this model of collaboration is not restricted to a single country.

Cifre PhD programme description

In this study, we decide to focus on a French PhD collaborative programme, the Cifre PhD programme; its name refers to the French Industrial Convention for Research Training. This programme is supported by the ANRT, a not-for-profit organization that acts on behalf of the French Ministry of Higher Education, Research, and Innovation. Founded in 1953, the ANRT has a mission to bring together all private and public stakeholders in French research in all sectors of activity and all scientific disciplines. In 2020, the ANRT had 354 members and partners (companies and public institutions), representing nearly three-quarters of national R&D efforts. Since 1981, the French Ministry of Higher Education, Research and Innovation (MESRI) has entrusted the ANRT with the implementation of the Cifre PhD programme (Industrial Agreement of Training through Research). The Cifre PhD programme provides financial assistance to French firms that recruit a PhD student to work on a research project for 3 years. The PhD candidate is therefore an employee of the firm while also being a student in a doctoral programme at a public university. He or she conducts his or her research through a university–industry collaboration under the supervision of a PhD supervisor attached to a public research laboratory and a scientific manager within the company. The doctoral student devotes 100% of his or her time, shared between the university and the company, to his or her research work.

According to the MESRI, between 1981 and 2017, nearly 9,000 companies, more than 4,000 academic laboratories and 25,400 doctoral students collaborated through the Cifre PhD programme. Each year, the Cifre PhD programme represents approximately 10% of all doctoral students in France. In 2020, 1,556 Cifre PhD projects were funded for

approximately 1,010 companies. For the ANRT, the objective of the Cifre PhD programme is twofold: first, to participate in the training of PhD students to support public and private research and, second, to promote the development of university–industry research collaborations based on knowledge creation.

The Cifre PhD programme provides a contractual link among the doctoral student, the company and the research laboratory (Figure 1). The doctoral student and the company are bound by a 3-year work contract. The doctoral student is enrolled at the public university to which the academic research laboratory is attached. As such, a thesis contract is established to specify the research project and the conditions for obtaining a doctoral degree from the university. Finally, the laboratory and the company are bound by a partnership agreement that specifies the terms of the collaboration—research project, intellectual property, exploitation of results, financial conditions, place of work of the doctoral student, etc.

[Insert Figure 1 near here]

Data collection

This research is based on the study of partnership agreements between the firm and the university (more precisely, the research laboratory). As our study started in 2019, we opt to analyse the Cifre PhD projects that were closed in 2018 (i.e., after a PhD duration of 3 years, as is traditional practice in France). Therefore, we are interested in the partnership agreements that were approved at the beginning of the projects in 2015. During this year, 1,377 projects were approved by the ANRT. With the approval of the ANRT, we select 904 partnership agreements, noting the distribution of large companies and SMEs. We cannot select all the partnership agreements for technical and timing reasons. As the partnership agreements are not archived in digital format, we face difficulty finding them

and have to scan them manually. We then selected only the projects that were carried out to their conclusion. In the context of our study, this indicates that the PhD student successfully defended his thesis and graduated. Indeed, we deliberately excluded the projects that were halted along the way, as confidentiality reasons made it impossible for us to know the causes of the collaboration's termination. For this purpose, we select only those projects for which the three parties (the doctoral student, the PhD supervisor, and the scientific manager from the firm) completed a document sent by the ANRT to certify that the projects had been completed and successful. We then eliminate the partnership agreements for which the technical appendix presenting the research project was missing or incomplete (15% of the sample). The final sample comprises 631 partnership agreements.

The university and the company are free to collaborate according to the respective expectations of the parties. They define, independently of the ANRT, the objectives of the research project, the various stages of the research, the organizational methods, the field of research, the scientific discipline, the expected results and the intellectual property. Partnership agreements are not standardized, but they generally contain the following information: time spent by the doctoral student between the company and the laboratory, financial provisions, intellectual property, exploitation of results and confidentiality conditions. In addition, a technical appendix to the partnership agreement specifies the PhD student's research project (between 1 and 10 pages), which is shared by the company and the academic research laboratory at the beginning of the collaboration.

Data analysis

For each research project, we analyse all partnership agreements to determine the research orientations and the nature of collaboration. In analysing partnership agreements, we are aware that we are studying the intention of the project and not its outcome. However, this situation corresponds perfectly to what we want to study: the nature of the research project

at the beginning of the university–industry collaboration. To evaluate research orientations, we analyse the technical appendices of the partnership agreements specifying the Cifre PhD research project. To do this, we use a hybrid process of inductive and deductive thematic analysis to interpret the raw data (Azungah, 2018; Fereday & Muir-Cochrane, 2006).

The inductive thematic analysis is based on detailed readings of the raw data (technical appendices of the partnership agreements) to generate categories and themes. We first analyse the data without preconceived categories or theoretical frameworks. We start by open-coding the data, which involves assigning descriptive or interpretive labels (codes) to segments of the data. These codes capture the essence of what is being expressed in the data without imposing preconceived categories. For example, we assign the label “*modelling*” to the technical appendix containing the following text: “*The aim of this research project is to provide a better understanding of the impact of the coupling between thermoelectric modules and their environment to optimize the performance at maximum power of such systems. For this purpose, a coupled approach between modelling (analytical and numerical) and characterization by impedance spectroscopy will be carried out on model systems (thermoelectric films and thermoelectric junctions) and then extended to more complex systems such as thermoelectric modules, which include several hundred junctions*” (Cifre PhD project #0861). As the coding progresses, we look for similarities, differences, and patterns in the coded data. We group similar labels together to form categories or themes. We identify nine labels and three different categories. This inductive coding is an iterative process, meaning that we continually refine and revise labels and categories as we delve deeper into the data and develop a more nuanced understanding of the phenomena under study (Figure 2).

[Insert Figure 2 near here]

Once initial patterns and themes are identified, we incorporate deductive coding to organize and analyse the data according to established concepts. The deductive labels are based on the research orientation categories proposed by Stokes (1997): *pure basic research*, *pure applied research* and *use-inspired basic research*. Stokes (1997) conceptualized Pasteur’s quadrant to categorize scientific research based on the following two dimensions: the quest for fundamental understanding and considerations of use. Based on this framework, the above author highlighted three kinds of research. *Pure basic research* seeks to enhance fundamental understanding without immediate consideration for practical applications. *Pure applied research* aims to solve specific problems or address practical challenges. *Use-inspired basic research* aims to create new fundamental scientific knowledge based on practical problems or identified application issues. This hybrid approach allows us to balance the flexibility of inductive coding with the structure provided by deductive coding, resulting in a comprehensive and nuanced analysis of the qualitative data (Table 1).

[Insert Table 1 near here]

The analysis of collaboration forms is based on the theoretical framework proposed by Lindl et al. (2013), which distinguishes four types of collaboration in knowledge creation partnerships: specified, distanced, developed, and translational. Due to the nature of our data, we excluded the transactional form of collaboration, as it did not apply to the research projects we studied. Specifically, within the Cifre programme, it was not feasible to manage two parallel projects. To assess the form of collaboration, we evaluated six criteria present in the collaboration contracts: (1) the time spent by the Cifre PhD student at the company and the research laboratory, (2) financial provisions, (3) intellectual property, (4) the exploitation of results, (5) confidentiality conditions and (6) prior

knowledge. Prior knowledge is the knowledge held by the laboratory and the company before the start of the research project, which may be necessary for its completion.

A specified form of collaboration was identified when a PhD student spends more than 70% of his or her time in the company, the company's financial contribution to the laboratory is minimal, prior knowledge is more critical for the company, and the intellectual assets are owned by the company, granting it exclusive rights to exploit the results. In this context, the industry emerges as the dominant actor in the collaboration. Conversely, the university is the dominant actor in the collaboration when the PhD student spends more than 70% of his or her time in the research laboratory, the company's financial contribution to the laboratory is significant, and the intellectual property and exploitation rights are shared equitably. In this case, we identified the collaboration as distanced. Finally, collaborations were considered developed collaborations when the PhD student spends more than 30% of his or her time at both institutions, and when the financial contributions, intellectual property, and prior knowledge are equitably shared.

Data analysis is carried out in the following three phases: the development of the coding methodology, the data coding and the harmonization of the data coding (Table 2). The first step consists of analysing approximately 100 Cifre partnership agreements from different years. This analysis allows us to explore the content of the partnership agreements, to test a first methodological approach to coding the different research orientations and to establish several categories and themes for each research orientation (Step 1 in Table 2). This dataset is excluded from our analysis to ensure consistency in the data and to reduce potential year effects.

[Insert Table 2 near here]

The second step focuses on coding the 631 partnership agreements beginning in 2015 (Step 2 in Table 2). During this phase, each contract is coded at least twice by two different people (double coding), and some partnership agreements are even triple coded. For this purpose, we rely on three research engineers. First, these research engineers are trained to recognize the three coding categories: basic research, applied research and use-inspired research. The first 40 partnership agreements are coded by the authors and serve as a form of basis for training the research engineers. Then, 160 partnership agreements are triple coded by two research engineers and one author. The results of the research engineers' coding are discussed in regular meetings with the authors to ensure the full ownership and harmonization of the methodology. Based on the comments of the research engineers, the list of project classification criteria is also updated in an iterative manner. The last phase continues with the coding of the 431 remaining partnership agreements (double coding).

Finally, the third step makes it possible to harmonize the coding discrepancies (approximately 30% of the 431 contracts). This harmonization is carried out in a collegial manner between the authors and the research engineers.

RESULTS

Research orientations in university–industry collaborations

Based on the qualitative coding of the technical appendices of Cifre partnership agreements, we categorized and evaluated the number of collaborations according to three research orientations: basic research, applied research, and use-inspired basic research (Table 3). We find that three-quarters of the collaborative research projects are based mainly on basic research (61%) or use-inspired basic research (11%). Only 28% of the research projects we examine can be qualified as applied research (research projects

explicitly linked to an industrial application, such as product and technology development).

[Insert Table 3 near here]

Although all collaborative research projects are sponsored by industry, in 61% of the cases, it is not possible to establish a direct link between the research objective and a potential industrial application. For example, as one Cifre partnership agreement states, *“The aim of this research project is to provide a better understanding of the impact of the coupling between thermoelectric modules and their environment to optimize the performance at maximum power of such systems. For this purpose, a coupled approach between modelling (analytical and numerical) and characterization by impedance spectroscopy is carried out on model systems (thermoelectric films and thermoelectric junctions) and then extended to more complex systems such as thermoelectric modules, which include several hundred junctions”* (Cifre PhD project #0861). In this case, the research orientation is considered basic research because the aim of the project is to create new fundamental scientific knowledge without considering applications and potential uses. Conversely, research projects categorized as applied research are explicitly linked to industrial applications. For example, one such project is described as a *“study of the phenomena contributing to the movement of a liquid solder and to the dissolution of separator sheets during the manufacture of heat exchangers”* (Cifre PhD project #1360); as another project states, *“The objective of this Cifre research project is to propose a strategy to identify with a high degree of certainty a user of a distance learning platform to deliver a certification attesting that the authenticated user has indeed followed and validated a training”* (Cifre PhD project #1126).

To better understand this distribution, we study the categories resulting from the inductive coding for each research orientation (Table 4). This precise analysis allows us to refine and enrich our understanding of the research orientations of collaborative research projects. We find that many basic research projects are based on modelling and simulation (47% of basic research projects and 29% of all Cifre projects studied). Academic research gap projects based on academic literature reviews represent 33% of basic research projects, and projects based on methodology issues represent 20%. Applied research projects are composed mainly of projects with the objectives of technology adaptation (47%) and prototype development (41%).

[Insert Table 4 near here]

This first series of findings highlights three main results. First, while previous research has solely exemplified the possibility of conducting basic research projects within the context of university–industry collaboration, our analysis goes a step forward and reveals that this mode accounts for more than half of the sampled projects. One reason for such a high share of cases might be linked to the fact that the creation of new knowledge through basic research is essential for making companies strategic decisions on future product lines and process technologies (Czarnitzki & Thorwarth, 2012; Griliches, 1985; Mansfield, 1980; Martinez-Senra et al., 2015; Nelson, 1959). Second, our study emphasizes that collaborative basic research includes mainly projects focusing on producing new scientific models and simulations, establishing new scientific methodologies, and generating novel insights by addressing identified knowledge gaps through thorough literature reviews. This empirical investigation also emphasizes the diverse and well-balanced objectives of collaborative research projects. We distinguish two research objectives representing more than 20% of the projects studied: modelling and simulation (basic research) and the

academic research gap (basic research). The following four research objectives each represent approximately 10% of the projects studied: methodology (basic research), technology adaptation (applied research), prototyping (applied research) and initial exploratory study (use-inspired basic research). Third, without denying the distinction between academic and industrial logic (Bjerregaard, 2010; Dasgupta and David, 1994; Sauermann and Stephan, 2013), this result emphasizes the facts that cultural distinctions between academic and corporate research are not necessarily obstacles to collaborative efforts aimed at creating new knowledge and that those collaborative projects mainly focus on exploring new scientific areas. In the context of research partnerships or collaborative research, collaborations tend to adopt the conventions of academic research and prioritize research orientations centred on basic research. While most studies have shown that collaborations based on technology transfer focus on applied research (Bozeman, 2000; Lee, 1996; O’Shea et al., 2005), our findings demonstrate that collaborations based on knowledge creation are oriented toward basic research. This result challenges and contradicts the notion that *“researchers working in organizations with a more applied research focus would favor the use of patents, labor mobility, and collaborative or contract research”* (Bekkers and Botas Freitas, 2008). Indeed, we demonstrate that collaborative research is primarily focused on basic research.

The influence of context on research orientations

To gain deeper insights into the range and diversity of research orientations within collaborative projects, we enhance our analysis by examining the relationships among research orientations, scientific disciplines, university types, and the nature of firms.

First, we find that research collaborations concern all scientific disciplines, from computer science to physics, including human and social sciences (Table 5). However, only

three disciplines represent 62% of all Cifre projects: computer science (26%), engineering sciences (21%) and chemistry/materials science (15%). Conversely, all collaborations based on mathematics, physics and life sciences account for less than 10% of Cifre PhD projects. While university–industry collaborations encompass all scientific disciplines, their distribution is not uniform. Like collaborations based on knowledge transfer, we find that those focused on knowledge creation also vary by discipline, with a higher frequency in the fields of chemistry, engineering, and information technology (Martinelli et al., 2008; Schartinger et al., 2002).

[Insert Table 5 near here]

However, we do not observe significant differences in research orientations across scientific disciplines. Basic research projects predominate in most fields. Except for physics, all disciplines are primarily oriented toward basic research: engineering sciences (75%), chemistry/materials science (67%), humanities (51%), health and medicine (70%), social sciences (91%), mathematics (53%), agricultural and biological sciences (61%), and life sciences (57%). Collaborations in computer science exhibit a balanced distribution, with 42% of projects focusing on basic research and 41% focusing on applied research. This finding suggests that, in the context of university–industry collaborations focusing on knowledge creation, scientific discipline does not significantly influence research orientation. Regardless of the scientific discipline, collaborations are primarily focused on knowledge creation projects with a basic research strategy. While previous studies emphasize that knowledge transfer channels depend on scientific disciplines (Martinelli et al., 2008; Schartinger et al., 2002), our findings present a new result: within a channel based on knowledge creation (collaborative research), scientific disciplines do not influence

research orientation. Consequently, the scientific discipline does not explain why collaborations primarily focus on basic research projects.

Second, we examine the relationship between research orientation and the type of university (Table 6). We observe that the type of university does not influence research orientation. For example, when comparing research-intensive universities and engineering schools that have extensively utilized the Cifre programme, we do not observe any differences. Despite their varied missions and institutional cultures, both research-intensive universities and engineering schools exhibit similar distributions of basic, applied, and use-inspired research orientations in their collaborations under the Cifre programme. This finding suggests that the collaborative research agenda is influenced more strongly by the mutual goals and negotiated interests of the university and industry partners than by the intrinsic characteristics or traditional roles of the institutions involved. Moreover, this result challenges studies suggesting university departments with a greater focus on applied research and technological development interact more intensively with industry (Agrawal, 2001; Lee, 1996, Bozeman, 2000; Colyvas et al., 2002; O’Shea et al., 2005). We do not refute this point, as most of these studies have focused on collaborations based on technology transfer. In contrast, our study highlights a specific case of collaboration based on knowledge creation. We demonstrate that, in this particular case, all types of academic institutions are capable of collaborating with industry on basic research projects.

[Insert Table 6 near here]

Third, we investigate the relationship between research orientation and the nature of firms. We assess the relationship between research orientations and company size (Table 7). Considering that university–industry collaborations are influenced by firm characteristics, particularly the company’s size and the intensity of its R&D activities

(Cohen et al., 2002; Fontana et al., 2006; Granstrand et al., 1997), we expect that smaller firms, with more limited R&D budgets than larger firms, are more likely to engage in collaborations with universities for applied research projects. However, we find that firm size has almost no impact on research orientation. Therefore, collaborative research between universities and small firms is not necessarily more oriented towards the development of concrete industrial applications. Therefore, basic research projects account for the majority of all projects, regardless of company size.

[Insert Table 7 near here]

This intriguing finding challenges the conventional expectation that smaller firms, constrained by limited R&D budgets, predominantly engage in applied research aimed at immediate industrial applications. This finding highlights the fact that even resource-limited firms recognize the strategic importance of engaging in fundamental research with the aim of remaining competitive and innovative in the long term.

We also analyse the influence of the company’s experience with the Cifre PhD programme (Table 8) on research orientation. When the company undertakes a research project through the Cifre PhD programme for the first time, we note that use-inspired basic research represents 30% of the research projects. This proportion is three times greater than that of a company that is accustomed to collaborating with a university through the Cifre PhD programme. We can therefore deduce that first-time users tend to turn to collaborative research when they are faced with exploration issues in the context of radical innovation. Among companies using Cifre PhD collaborations occasionally, regularly or systematically, no significant differences can be observed.

[Insert Table 8 near here]

This finding suggests that new entrants engage in this type of research with the goals of exploring and addressing radical scientific challenges, thereby combining foundational inquiry with practical relevance. In contrast, experienced collaborators tend to feature a more balanced distribution of research orientations, likely because they have already established a foundational knowledge base and can now integrate both basic and applied scientific research more effectively. This evolution highlights the dynamic nature of university–industry collaborations, in which context firms adjust their research strategies to optimize scientific knowledge creation as they gain more experience with collaborative projects.

We note that many previous works have focused on different collaboration channels (patent and licensing, collaborative research, informal contact, and labour mobility) according to context (scientific discipline, industrial sector, and company size). For example, Santoro and Chakrabarti (2002) showed that large firms more often resort to knowledge transfer, whereas small firms prefer cooperative research-based relationships. Many previous empirical investigations have shown that larger firms have a much greater probability of R&D collaboration with universities than do small firms (Arundel & Geuna, 2004; Cohen et al., 2002; Laursen & Salter, 2003; Mohen & Hoareau, 2003). Fontana et al. (2006) even showed that firm R&D intensity affects both the propensity for and the extent of engagement in university–industry collaborations. Without contradicting these results, our findings provide new details and contributions by focusing exclusively on collaborations based on knowledge creation. Our aim is not to compare knowledge transfer channels but to study the objectives and expectations pursued in collaborative research projects, an aspect that has not been highlighted in previous contributions. Our findings suggest that research orientations are not affected by scientific discipline, university type or the nature of firms, meaning that the nature of knowledge creation (knowledge creation to address a practical problem or knowledge creation driven by a quest for understanding) does not

depend on exogenous factors but only on the objectives and choices of both partners—the university and the industry.

Research orientations based on the forms of collaboration

According to Lindl et al. (2013), university–industry collaboration within research partnerships can take various forms. Collaboration based on knowledge creation can be dominated by the university logic, the industry logic, or both. Consequently, we sought to investigate the relationship between the research orientations of the projects and the forms of collaboration. As explained in the methodology section, we identified three types of collaboration based on six elements present in the collaboration agreement: (1) distanced collaboration (i.e., university logic), (2) specified collaboration (i.e., industry logic); and (3) developed collaboration (i.e., both university and industry logics are influential)

In the context of Cifre research partnerships, our results reveal a balanced distribution of the three different forms of collaboration (Table 9). In 39% of cases, research projects follow a developed form of collaboration, 33% follow a distanced form, and 28% are based on a specified form of collaboration. In most cases (72%), the university laboratory plays a key role in managing the research projects, acting as the leader in 33% of cases and as coleader in 39%.

[Insert Table 9 near here]

We then analysed the forms of collaboration by company type (Table 10). Our findings indicate that the form of collaboration does not depend on the type of company, as different forms of collaboration are balanced across company sizes. Developed collaborations are the most common form for startups, SMEs, and mid-sized companies. In contrast, large firms appear to be better equipped to lead collaborative research projects

than smaller firms. This can be attributed to the fact that large firms often have R&D departments with highly qualified technical and scientific staff (Fontana et al., 2006).

[Insert Table 10 near here]

We also analysed the relationships between research orientations and forms of collaboration (Table 11). Based on this analysis, we identify three main findings. First, in the context of collaborations based on knowledge creation, we found that the form of collaboration does not influence the research orientation. Indeed, regardless of the research orientation, the different forms of collaboration are present in a balanced manner. For example, in the case of basic research, there are as many projects based on distanced collaborations (23%) as on developed collaborations (23%). We observe the same in the case of applied research. There are almost as many projects based on distanced collaborations as there are on developed and specified collaborations. Second, we emphasize that the modalities and strategies of collaboration aimed at creating new knowledge are varied, diverse, and heterogeneous. Indeed, these projects can be organized according to at least nine ideal types of university–industry collaboration. For example, projects based on a distanced form of collaboration can pursue three distinct research strategies: basic research, applied research, or use-inspired basic research. This means that university–industry collaborations based on knowledge creation can be applied in a wide variety of contexts without compromising the performance of these collaborations. Third, we observe that the distribution of the nine ideal types of university–industry collaboration is not fully balanced. Instead, two types dominate, representing 46% of the sample: basic research conducted through distanced collaboration (23%) and basic research through developed collaboration (23%). Therefore, while collaborations can be organized in many ways, both

substantively and structurally, in the context of knowledge creation research partnerships, the university logic appears to be the most influential.

[Insert Table 11 near here]

DISCUSSION

This study aimed to explore the nature of research orientations in university–industry collaborations, specifically in the context of the Cifre PhD programme. Following the literature review, we aimed at answering the following research questions: (1) What is the nature of research orientations in university–industry collaborations focused on knowledge creation? (2) What is the influence of context on research orientations? (3) What are the relationships between research orientations and forms of collaboration? Based on an analysis of 631 partnership agreements, we categorized the research orientations as basic research, applied research, and use-inspired basic research, we examined the influence of context on research orientations and the influence of the forms of collaborations on research orientations. We have shown that more than half of the projects (61%) focus on basic research, highlighting a strong commitment by industry to fundamental scientific inquiry. While industry sponsors joint research projects, these findings indicate that research collaborations mainly embrace academic research conventions, the university logic and prioritize orientations focused on basic research. This trend was consistent across different scientific disciplines, university types, and firm sizes, thus challenging the claim that industry-driven projects predominantly focus on applied research (Bozeman, 2000; Lee, 1996; O’Shea et al., 2005). These findings highlight the strategic value of basic research in industry-sponsored projects as well as the importance of developed collaborations. Our results provide a comprehensive overview of the diversity and dynamics of these

partnerships, paving the way for future research and policy development aimed at enhancing university–industry collaborations.

Theoretical and empirical contributions

Our study contributes to the literature on university–industry collaboration in several ways. First, we emphasize the significance of focusing on research orientations during the negotiation phase of collaborative projects. Existing studies on university–industry collaboration often concentrate on the performance and success of such projects (Bikard et al., 2019; Gulbrandsen & Smeby, 2005; Perkmann & Walsh, 2009; Perkmann et al., 2021). Conversely, our study focuses on the preliminary phase of the collaboration process, specifically the decisions and motivations that precede the initiation of various collaborative research projects on the basis of knowledge creation (Hottenrott, 2012; Ruoslahti, 2020). Our study indicates that those collaborations are oriented mainly towards research orientations focusing on the creation of new scientific knowledge without considering potential applications and uses. Such cases are not only rare events where both parties managed to align their goals and expectations toward fundamental inquiries; on the contrary, they constitute the majority of the studied projects. In other words, despite the industry's bargaining power, projects are not focused mainly on developing knowledge for short-term integration in the new product development process. Instead, companies are more willing to collaborate with universities to nourish their knowledge base with knowledge that can help them sustain their R&D process over the long run.

Second, our study contributes to understanding university–industry collaborations by revealing that the traditional influence of contextual factors, such as scientific discipline, university type, and firm characteristics, on research orientations is not as significant as expected. Despite the wide range of scientific disciplines involved in these collaborations, most projects, including those in traditionally applied fields such as engineering and

computer science, remain focused on basic research. This challenges the assumption that certain disciplines inherently lean towards applied research (e.g., Martinelli et al., 2008; Schartinger et al., 2002). Additionally, we find that the type of academic institution—whether a research-intensive university or an engineering school—does not significantly affect the research orientation of collaborative projects. Both types of institutions exhibit similar distributions of basic, applied, and use-inspired research, suggesting that the research agenda is shaped more by the mutual goals of the university and industry partners than by the intrinsic characteristics of the institutions themselves. Moreover, our findings show that firm size, traditionally thought to influence a preference for applied research due to resource limitations (e.g., Fontana et al., 2006), has little impact on research orientation. Both small and large firms engage predominantly in basic research, indicating that even smaller firms recognize the strategic importance of investing in foundational knowledge for long-term competitiveness. Overall, these results suggest that the typical contextual influences seen in university–industry collaborations are less relevant in projects focused on knowledge creation, where research orientation is driven primarily by the specific goals and negotiated interests of the partners involved.

Third, our study contributes to the literature on the asymmetry between industry and academia research orientations and the manner in which it constitutes a barrier for those collaborations (Atta-Owusu et al., 2021; Bjerregaard, 2010; Callaert et al., 2015; He et al., 2021; Perkmann et al., 2013; Sauermann and Stephan, 2013; York et al., 2016). Our results reveal that those asymmetries can be overcome by the possibility of adopting very different settings and research orientations in those collaborative research projects. Given that collaborations often align with the practices of university research, this exploratory study also addresses the concern raised by Bruneel et al. (2010) regarding the insufficient comprehension of the dynamics underlying such collaborations.

More generally, our study contributes to the literature on research and innovation policy. Our empirical results confirm the limitations of the conceptual linear model of innovation (Calvert, 2006; Cantisani, 2006; Goldstein & Narayanamurti, 2018; Gulbrandsen & Kyvik, 2010; Narayanamurti et al., 2013; Tijssen, 2010). Our findings show that industry-supported research projects are not limited to applied research. This finding does not invalidate the fact that basic research can be conducted independently of any industrial application considerations. However, our findings highlight that there is no strict or impermeable boundary between, on the one hand, basic research financed and managed exclusively by public laboratories and, on the other hand, applied research financed and organized exclusively by the private sector. In contrast, our findings show that university–industry research collaborations facilitate the creation of new scientific knowledge without preexisting industrial applications. Our study also highlights a university–industry collaboration framework that is well-suited to Mode 2 of knowledge production. While technology transfer appeared as the dominant system in Mode 1, our study underscores the relevance of collaboration based on knowledge creation within the context of Mode 2. This result also contributes to the research on how private firms perform and exploit basic research with their own money (Pavitt, 1911; Rosenberg, 1990; Salter & Martin, 2001). Firms conduct basic research to build a knowledge base that provides them with the fundamental basis for subsequent innovation and technology development. Basic research enables a firm to exploit rapidly useful scientific and technological knowledge through its own innovations. Given that this strategy is driven by scientific questions and is phenomenon oriented, collaboration with universities and research institutes is essential. These collaborations support the absorptive capacity to use external knowledge and stimulate fresh perspectives, potentially resulting in the generation of innovative and novel ideas, concepts, and applications (Arora et al., 2017; Belderbos et al., 2010; Czarnitzki & Thorwarth, 2012). To bridge the gap between basic and applied research, it is essential to

separate scientific questioning, research orientation, and the practical utility of research. Industrial issues can generate research questions (scientific questioning) and produce fundamental knowledge (research orientation), which can then be used for concrete applications (utility of research). Thus, the identification of concrete industrial applications is not a prerequisite for the development of joint research projects inspired by or originating from industry. Finally, the concept of applied research itself may be confusing. Like Pasteur (1939), we defend the idea that scientific research activity is primarily a process of producing theoretical knowledge controlled by the scientific method. Consequently, such activity can in no way be confused with the production of knowledge from practice, which is explained as follows: *“There is no such thing as applied science. The very union of these words is shocking. However, there are applications of science, which is quite different”* (Pasteur, 1939, p. 186).

Practical implications

Our findings have implications for the management of university–industry collaborations as well as for research and innovation policies.

The apparent distinction between academic logic and commercial logic suggests that collaboration on the basis of knowledge creation is not easy. The lack of alignment in research orientations can potentially generate contradictory objectives and expectations regarding various aspects of such a collaborative research project. First, our findings reveal that companies and universities can collaborate by codesigning a common research programme and agreeing on a shared research orientation, thus indicating that collaborations based on knowledge creation are complementary to collaborations based on knowledge transfer. However, participants in collaborative projects should concentrate on joint endeavours grounded in basic research. Our results indicate that collaborations often adhere to the practices of academic research and prioritize research orientations centred on basic research. Furthermore, the results highlight different basic research objectives,

such as scientific modelling and simulation, establishing new scientific methodologies, and generating novel insights by addressing identified knowledge gaps through thorough literature reviews. Second, our findings suggest that research orientations are not subject to scientific disciplines, types of universities, or types of companies. In the context of Cifre PhD collaborations, partners can participate in basic research projects regardless of the scientific discipline or the size of the company. The key challenge lies in how to establish clear objectives and expectations for collaboration at the outset. We believe that this principle applies broadly to all collaborations involving industrial PhD programmes (Abu Sa'a & Yström, 2024; Borrel-Damian et al., 2015; Grimm, 2018; Salimi et al., 2016).

In the context of innovation competition and the knowledge-based economy, academic and research institutions are increasingly encouraged to expand their missions to contribute more to the society on whose behalf they work, which is why universities engage in technology transfer activities to commercialize the results of their research and innovations. This situation has resulted in the licensing of intellectual property rights, the creation of spin-off companies or the establishment of partnerships with industry to develop and commercialize new products, processes and technologies (Chai & Shih, 2016; Rajalo & Vadi, 2017; Siegel et al., 2003; Terán-Bustamante et al., 2021). Our findings suggest the feasibility of fostering collaborations that go beyond simply commercializing research outcomes. Collaboration based on knowledge creation enables the production of new scientific knowledge that may be beneficial to both academia and industry. To accomplish this goal, university–industry collaborations must go beyond the strict dichotomy between basic and applied research and must move away from the linear model of innovation, which presupposes a division of labour between researchers in universities and public laboratories, with a focus on basic research and researchers in corporate R&D departments focusing on applied research (Balconi et al., 2010). Technology can even become a source of questioning for basic research, as it highlights complex problems,

dilemmas, and needs for instrumentation (Calvert, 2006; Cantisani, 2006; Goldstein & Narayanamurti, 2018; Gulbrandsen & Kyvik, 2010; Narayanamurti et al., 2013; Tijssen, 2010). One of the best-known examples is the development of the second law of thermodynamics in the 1820s by Sadi Carnot, which was motivated by the need to understand the efficiency limits of steam engines (Balconi et al., 2010). Therefore, innovation is not a linear, deterministic process that is composed of sequences of independent steps, with no possible feedback among basic research, applied research, technological development, and product commercialization (Clarke, 2010; Henard & McFadyen, 2005; Jansen, 1995; Walsh, 1984).

Limitations and future research directions

Because of the exploratory nature of this study, several limitations should be considered. First, due to the typically small and nonrepresentative samples used in exploratory research, the findings may not be generalizable to larger populations or contexts. Indeed, we decide to focus on university–industry collaborations on the basis of PhD collaborative programmes, also called industrial PhDs (Abu Sa’a & Yström, 2024; Grimm, 2018; Salimi et al., 2016). In particular, we study a French PhD collaborative programme—the Cifre PhD programme—and focus on 631 partnership agreements associated with the same year. Hence, it would be preferable to supplement this study with data concerning other PhD programmes in diverse countries. Second, this research may be affected by several biases due to subjective interpretation or selective attention to the data, particularly because it is a qualitative exploratory study. Additional empirical research could enhance and expand upon this initial approach to the task of analysing knowledge creation in the context of university–industry collaborations. Third, this exploratory research focuses on understanding relationships and patterns and does not establish causality among variables. Our findings suggest that research orientations are not affected by scientific discipline, university type or the nature of firms but cannot determine cause-and-effect relationships.

This exploratory study offers a general perspective and an initial understanding of research orientations in university–industry collaborations centred on knowledge creation, but it may not delve deeply into specific aspects or variables. Furthermore, by examining the creation phase of collaboration without making assumptions regarding outcome, this research does not make it possible to evaluate the performance and outcomes of collaborations according to research orientation. One potential avenue for future research could be to explore the performance of research collaborations on the basis of their research orientations. For example, it might be interesting to analyse patents and scientific publications both quantitatively and qualitatively. Additionally, it could be valuable to study the influence of research orientations on the originality of research projects, particularly by examining whether the emergence of breakthrough innovations is influenced by such orientations. It would also be interesting to determine whether university–industry collaborations facilitate more exploratory research by entering into new fields and addressing new problems, which could result in a greater diversity of research impacts. Fourth, given that collaborative PhD programmes represent only one mechanism of university–industry collaboration, among other possibilities, it would be desirable to explore the emergence of university–industry collaborations within other frameworks. For example, future research could analyse research orientations in research consortia and joint research centres. Research consortia bring together multiple organizations, such as universities, research institutes, and industry partners, to collaborate on specific research projects or themes. Joint research centres are established partnerships between academic institutions and industry or government entities to conduct research in a particular field or area of interest. Both are collaborative arrangements that extend beyond mere knowledge transfer, aiming instead to generate new scientific knowledge. Consequently, to enhance our understanding of the influence of research orientations in the context of university–industry collaborations, we encourage future

researchers to conduct comparative studies. These studies could involve comparing several factors, such as collaborative frameworks, scientific fields, company sizes, and levels of collaborative experience, with the goal of identifying similarities and differences. International comparative studies would also provide us with a better understanding of the performance, effectiveness, and characteristics of collaborations in relation to research orientations.

Therefore, we synthesize a comprehensive research agenda with the goal of enhancing our understanding of university–industry collaborations, particularly from the perspective of research orientation (Table 12). This agenda identifies key themes and subthemes that require further exploration to elucidate the complexities and dynamics underlying these collaborations. Specifically, the agenda focuses on the roles of research orientation, industry and academic engagement, strategic approaches to collaboration, and the implications of research policies. By investigating these areas, we aim to reveal the nuanced factors that drive successful and sustainable collaborations, the varying motivations and outcomes for different types of research engagement involved in this context, and the impact of organizational and policy frameworks on these partnerships. This research agenda provides a structured framework that can guide future studies by proposing a set of pertinent research questions that address both theoretical and practical aspects of university–industry collaborations across diverse scientific disciplines and international contexts.

[Insert Table 12 near here]

CONCLUSIONS

This study investigates university–industry collaborations within the Cifre PhD program, with a focus on research orientations. An analysis of 631 partnership agreements revealed

that basic research is predominant, and the choice of this research orientation is not contingent on specific contextual factors, thus challenging the expectation that industry-driven projects are mainly applied. These findings highlight the strategic value of basic research with respect to long-term innovation and competitiveness. They also suggest that fostering collaborations that integrate both basic and applied research can promote technological and societal advancements. The proposed research agenda encourages further exploration of the factors influencing research orientations, the impact of early-stage industry engagement, and the role of policies in shaping collaboration outcomes. Understanding and strategically managing research orientations can enhance university–industry collaboration, thereby ultimately contributing to innovation and addressing global challenges.

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FIGURE LEGENDS

Figure 1. Structure of the Cifre PhD programme (based on ANRT documentation)

Figure 2. Coding process

TABLES

Table 1. Inductive and deductive thematic analysis

Inductive coding		Deductive coding	
Labels	Description	Labels	Description
Modelling & simulation	Research project focused on the development of models or on simulation design characteristics	Pure basic research	The research project aims to create new fundamental scientific knowledge without considering applications and potential uses.
Academic research gap	Research project focused on a research gap based on an academic literature review		
Methodology	Research project focused on methodology issues		
Technology adaptation/improvement	Research project focused on technology adaptation or improvement	Pure applied research	The research project aims to create scientific knowledge directly applied to the identified product, process or service of the organization.
Prototyping	Research project focused on the development of a prototype		
Product optimization	Research project focused on the optimization or improvement of a product		
Process and Service design	Research focused on the development of a process or a service		
Initial exploratory study	Research focused on explorative studies in a radical innovation context	Use-inspired basic research	The research project aims to create new fundamental scientific knowledge based on a practical problem or an identified application issue
Follow-up exploratory study	Research focused on the continuation of exploratory studies		

Table 2. Data analysis process

Data analysis step	Data	Data coding process/objectives
Step 1: Development of the coding methodology	100 partnership agreements (excluded from the dataset)	Double coding (authors) Partnership agreement analysis and development of the coding methodology
Step 2: Data coding	40 partnership agreements	Triple coding (2 research engineers and one author)/training of research engineers for data coding
	160 partnership agreements	Triple coding (2 research engineers and one author)
	431 partnership agreements	Double coding (2 research engineers) 4 milestones
Step 3: Harmonization		Analysis of coding discrepancies ($\approx 30\%$) and correction

Table 3. Research orientations of the Cifre PhD programme

Research orientation	Number of Cifre PhD projects
Basic research	384 (61%)
Applied research	175 (28%)
Use-inspired basic research	72 (11%)
Total	631 (100%)

Table 4. Distribution of Cifre PhD projects by subcategory of research orientation

Research orientation	Subcategories based on inductive coding	Cifre project	PhD project (%)
Basic research 384 projects (61%)	Modelling & simulation	180 (47%*)	29%
	Academic research gap	130 (33%*)	20%
	Methodology	74 (20%*)	12%
	Technological adaptation	83 (47%*)	13%
Applied research 175 project (28%)	Prototyping	72 (41%*)	11%
	Product optimization	15 (9%*)	2%
	Services design	5 (3%*)	1%
Use-inspired basic research 72 projects (11%)	Initial exploratory study	60 (83%*)	10%
	Follow-up exploratory study	12 (17%*)	3%

*% within each research orientation

Table 5. Research orientations according to scientific discipline

	Basic research	Applied research	Use-inspired basic research	Total
Computer sciences	70 (42%)	69 (41%)	27 (17%)	166 (26%)
Engineering sciences*	98 (75%)	28 (22%)	4 (3%)	130 (21%)
Chemistry/materials science	63 (68%)	22 (24%)	8 (9%)	93 (15%)
Humanities**	31 (51%)	23 (38%)	7 (11%)	61 (10%)
Health and medicine	39 (70%)	9 (16%)	8 (14%)	56 (9%)
Social sciences***	40 (91%)	1 (2%)	3 (7%)	44 (7%)
Mathematics	19 (54%)	8 (23%)	8 (23%)	35 (6%)
Agricultural and biological sciences	17 (61%)	7 (25%)	4 (14%)	28 (4%)
Physics	3 (27%)	6 (55%)	2 (18%)	11 (2%)
Life sciences	4 (57%)	2 (29%)	1 (14%)	7 (1%)

*Mechanics, electronics, and automation; **language sciences, history, geography, education sciences, psychology, and urban planning; and ***law, economics, management, sociology, and political science.

Table 6. Research orientation by type of university

University			Basic research	Applied research	Use-inspired basic research
Type of university					
Aix Marseille University	Research-intensive university		43%	30%	26%
Sorbonne University	Research-intensive university		64%	23%	14%
INP Grenoble University	School of engineering		55%	36%	9%
Bordeaux University	of Research-intensive university		81%	14%	5%
University of Montpellier	Research-intensive university		45%	40%	15%
University of Toulouse	Research-intensive university		74%	21%	5%
CentraleSupélec	School of engineering		59%	29%	12%
INSA Lyon	School of engineering		85%	15%	0%
Ecole Centrale Lyon	School of engineering		64%	36%	0%

postprint version (accepted author manuscript), to cite the article:

Cabanes, B., Plantec, Q., Le Masson, P. & Weil, B. (2024) Basic or applied research in university–industry collaborations? An analysis of research orientations in knowledge creation partnerships. *European Management Review*, 1–22. <https://doi.org/10.1111/emre.12696>

Table 7. Research orientation by type of company

	Basic research	Applied research	Use-inspired basic research	Total
Startups	36 (51%)	28 (40%)	6 (9%)	70 (11%)
SMEs	114 (61%)	52 (28%)	20 (11%)	186 (29%)
Mid-sized companies	74 (65%)	25 (22%)	14 (12%)	113 (18%)
Large companies	160 (61%)	70 (27%)	32 (12%)	262 (42%)

Table 8. Research orientations and company experience with the Cifre PhD programme

	Basic research	Applied research	Use-inspired basic research	Total
First university–industry collaboration based on the Cifre PhD programme	27 (47%)	13 (23%)	17 (30%)	57 (9%)
Occasional university–industry collaborations through the Cifre PhD programme	59 (64%)	25 (27%)	8 (9%)	92 (14%)
Regular university–industry collaborations through the Cifre PhD programme (one ongoing project)	64 (65%)	26 (26%)	8 (8%)	98 (16%)
Very regular university–industry collaborations through the Cifre PhD programme (several ongoing projects)	234 (61%)	111 (29%)	39 (10%)	384 (61%)

Table 9. Nature of collaboration in Cifre PhD collaborative projects

Nature of collaboration	Number of Cifre PhD projects
Distanced collaboration	208 (33%)
Specified collaboration	179 (28%)
Developed collaboration	244 (39%)
Total	631 (100%)

Table 10. Nature of collaboration by type of company

	Distanced collaboration	Specified collaboration	Developed collaboration	Total
Startups	18 (26%)	20 (29%)	32 (46%)	70 (11%)
SMEs	61 (33%)	39 (21%)	86 (46%)	186 (29%)
Mid-sized companies	41 (36%)	28 (25%)	44 (39%)	113 (18%)
Large companies	88 (34%)	92 (35%)	82 (31%)	262 (42%)

Table 11. Research orientations and forms of collaboration

	Distanced collaboration	Specified collaboration	Developed collaboration
Basic Research	143 (23%)	94 (15%)	147 (23%)
Applied Research	45 (7%)	60 (10%)	70 (11%)
Use-inspired Basic Research	20 (3%)	25 (4%)	27 (4%)

Table 12. Research agenda

Theme	Opportunities for future research	Possible research questions
Role of research orientation in university–industry collaborations	Investigate the impacts of different research orientations on the outcomes of collaboration.	- How do different research orientations (basic, applied, and use-inspired) affect the success and sustainability of collaborations? - What processes are used by universities and companies to determine their choice of research orientation in the context of collaborations?
Industrial/academic engagement in university–industry collaborations	Examine the motivations and outcomes associated with industry participation in basic research.	- How does basic research influence innovation capabilities and market competitiveness? - What challenges and benefits are entailed by industry engagement in basic research?
	Study variations in research orientations and collaboration types across different scientific disciplines.	- Why do research orientations and collaboration structures vary across disciplines? - What role does research orientation play for different groups of discipline (e.g., social sciences vs. STEM disciplines)? - How does interdisciplinary collaboration affect research outcomes?

Strategizing university–industry collaborations	Investigate the influence of different types of universities and companies on collaboration dynamics.	- How does company size (startups, SMEs, large firms) impact collaboration dynamics and research outcomes and depend on research orientations (e.g., processes)? - How are SMEs and startups targeted by policy-makers to foster university–industry collaborations?
	Study the exploratory nature of early-stage industry engagement in academic research.	- How do early-stage collaborations influence the direction and success of subsequent research projects? - What factors encourage actors in industry to engage in exploratory academic research through university–industry collaborations? - How do early-stage collaborations foster innovation and reveal new research areas?
Research policies regarding university–industry collaborations	Assess the implications of research orientations on policy and management practices.	- What policies can enhance the effectiveness of university–industry collaborations based on different research orientations? - How can management practices be adapted to support diverse research orientations? - What are the best practices for aligning academic and industrial research goals?

	Compare research orientations and collaboration outcomes across different countries.	- How do research orientations and collaboration structures in the Cifre PhD programme compare to those of similar programs in other countries? - How do national policies influence the success of university–industry collaborations?
	Investigate the long-term impacts of different research orientations on technological and societal advancements.	- What is the long-term impact of basic vs. applied research orientations on technological advancements? - How do university–industry collaborations help address societal challenges through different research orientations?

postprint version (accepted author manuscript), to cite the article:

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FIGURES

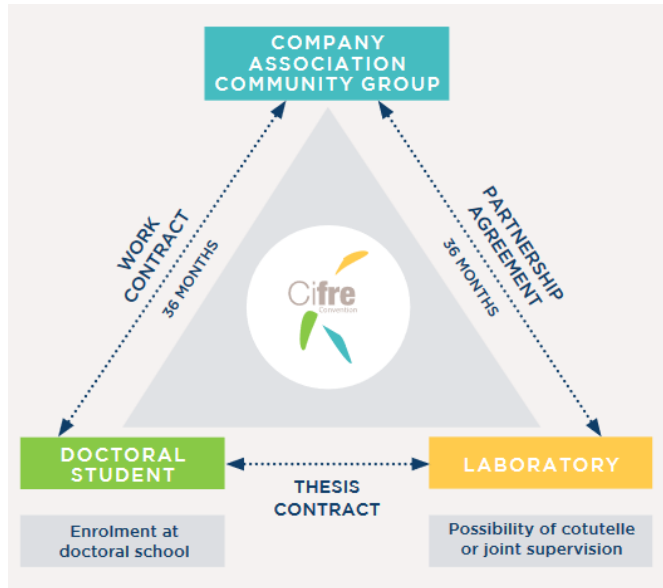


Figure 1

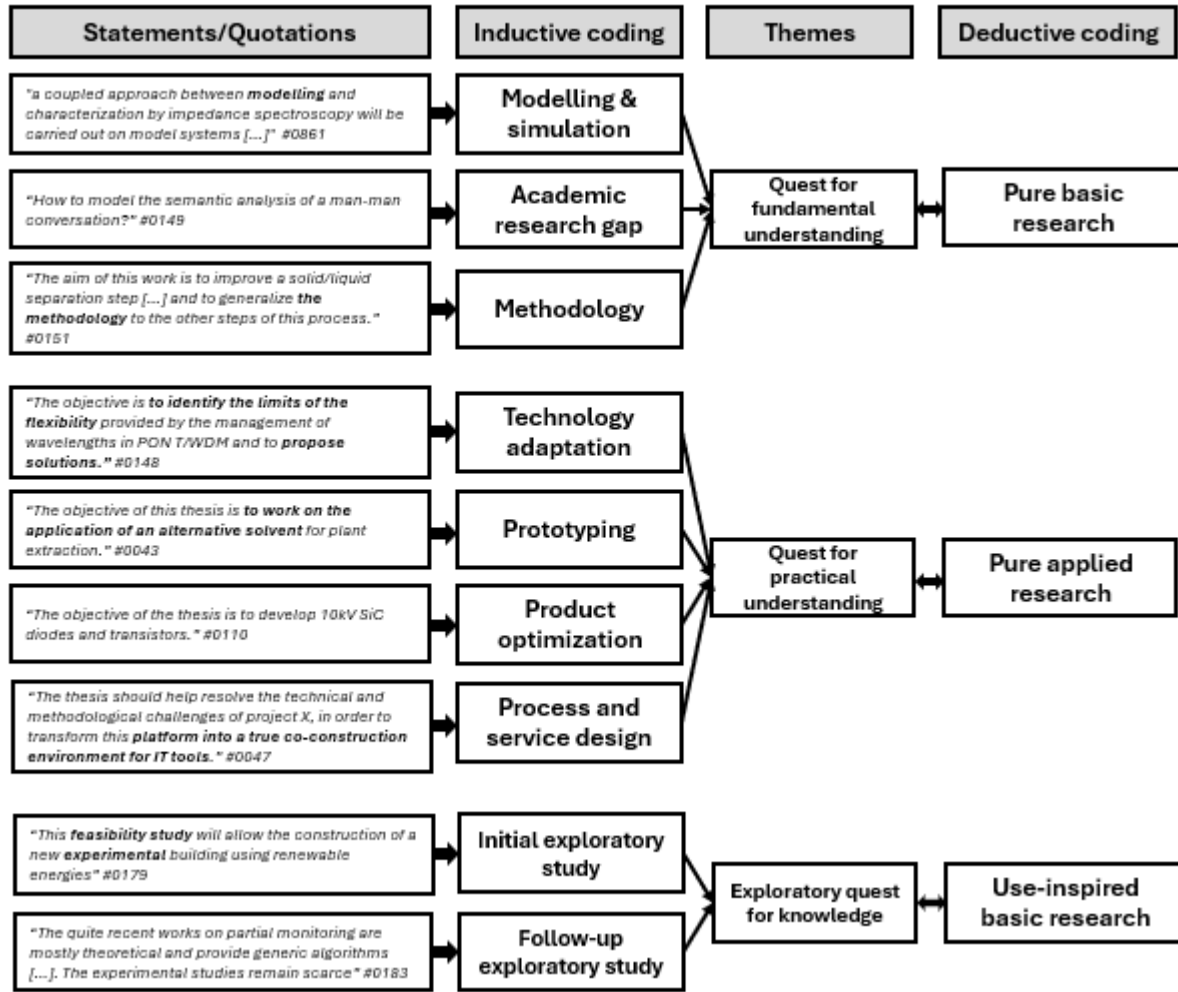


Figure 2. Coding process